IEEE Paper Viva Preparation Guide

Legal Document Clustering via TF-IDF, UMAP & Multi-Algorithm Comparison

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Dataset	1,000 Australian Federal Court Cases (Kaggle)
Language / Tools	R 4.5.3 | TF-IDF | UMAP | K-Means | HDBSCAN | Hierarchical Clustering
Guide prepared	For Today's Viva / Defense — May 2026

Tip for Viva: Ei guide-er key numbers gulo (Silhouette Score, DB Index, improvement %) valo kore mone rekho. Viva te numbers diye answer korle examiner impressed hoy!

PART 1 — Paper-er Basic Concept: Ekdom Shohoz Banglay

1.1 Paper-ta Mota-Muti Ki Niye?

Ei paper-e amar team **1,000 ti Australian Federal Court-er legal case document** niyeche ebong seigulo ke automatically group kore felar (cluster) cheshta koreche — **kono manual labeling baa supervised training chara**. Eita holo **Unsupervised Machine Learning**.

S■■ udaharan: Manush jodi 1,000 ta document pore manually group kore, time lage onek. Amader system ta automatic bhabe bole dite pare — 'Ei 200ta case immigration niye, oi 200ta tax niye, ar oi 200ta intellectual property niye.'

> ■ **Ekline Summary:** Legal documents ke automatically category-te bhaag kora — kono human expert chara — shudhumatro text analysis diye.

1.2 Keno Ei Kaajtai Dorkar Chilo?

- **Manual review expensive:** Legal annotator cost kore $150–$350 per hour. 1,000 case manually sort kora practically impossible.
- **Labeled data nai:** Court cases-er kono pre-made categories thake na — supervised learning kaj kore na.
- **Variety:** Immigration, tax, IP, competition — onek rokomer case thake ek sathe.
- **Speed:** Amader system 1,000 document-ke 2 second-er kam time-e cluster kore feltel pare.

1.3 Paper-er Main Pipeline — 5 Step-e

Step 1	**Data Collection** Kaggle theke 1,000 Australian court case download koro — 5 category theke 200 kore balanced sample
Step 2	**Text Preprocessing** Raw text ke clean koro — lowercase, punctuation remove, stopwords remove, stemming (7 steps)
Step 3	**TF-IDF Vectorization** Har document ke 1,175 number-er ek row-te convert koro (mathematical representation)
Step 4	**UMAP Reduction** 1,175 dimension theke shudhu 2 dimension-e compress koro (scatter plot-er moto)
Step 5	**Clustering + Evaluation** K-Means, HDBSCAN, Hierarchical — tin algorithm diye group koro, tahole measure koro quality

PART 2 — Pipeline: Step-by-Step Explanation

2.1 Dataset — Data Kothay Theke Aslo?

Source: Kaggle Legal Text Classification Dataset — Australia-r real Federal Court cases.

Full dataset: 24,985 cases across 10 categories. **Amra use korechhi:** 1,000 cases (5 category × 200 each).

Category	Ki Bujhay?	Count	Avg Chars
Cited	Shudhu reference — kono endorsement nai	200	2,847
Referred To	Contextually relevant bole mention	200	2,611
Applied	Legal principle directly apply kora hoyeche	200	2,534
Followed	Court agree koreche and follow koreche	200	2,490
Considered	Carefully examine kora hoyeche	200	2,712

2.2 Text Preprocessing — 7 Steps

Raw text mane ganda, mixed data. Algorithm-ke feed korar age sheta clean korte hoy. Amra exactly 7-ta step use korechhi:

1	**Lowercasing** 'Court' ar 'court' ke same treat kora
2	**Punctuation Remove** All . , ; : ( ) remove — vocabulary fragmentation kame
3	**Number Remove** Case numbers, dates, citation numbers — semantic value nai, remove
4	**Standard Stopwords** 174 ta English stopword remove (the, a, is, was…)
5	**Legal Domain Stopwords** 28 ta legal-specific word remove (shall, court, judgment, may…)
6	**Porter Stemming** agreements → agre, immigration → immigr, tribunal → tribun
7	**Whitespace Normalize** Multiple spaces → single space

2.3 TF-IDF Vectorization — Document ke Number-e Convert

TF-IDF mane ki? Term Frequency-Inverse Document Frequency. Ei method bole — kono word ki sei specific document-e beshi ache but overall corpus-e rare? Sei word-er importance beshi.

Formula (shohoj bhashay):
TF = Kono word ki sei document-e kotobar ache / Total words in document
IDF = log(Total documents / Documents containing that word)
TF-IDF = TF × IDF — word ta koto important sei document-er jonno

Result: 1,000 document × 1,175 terms = ek bora matrix (1,000 rows, 1,175 columns). Each row = ek document. Each column = ek word-er importance score.

2.4 UMAP — 1,175 Dimension theke 2 Dimension-e!

Problem: 1,175 column niye cluster kora kaaj korena — 'Curse of Dimensionality'. High dimension-e shob distance almost same hoye jay — cluster distinguish kora jaye na.

Solution — UMAP: Uniform Manifold Approximation and Projection. Ei algorithm 1,175D data ke 2D-te compress kore — ebhabe je similar documents kache kache thake ar different gulo dur dur thake.

Imagine koro: 1,175 dimension = 1,175 axis diye ek bora space. UMAP seita flatten kore ek 2D scatter plot banie fel■ — jeta tumi screen-e dekhte parao. Immigration cases ek corner-e, Tax cases ek corner-e — visually clear!

Parameter	Value	Keno?
n_neighbors	15	Local ar global structure-er balance
n_components	2	2D output for visualization
metric	cosine	Document length-er effect avoid kore
min_dist	0.05	Clusters compact hobe
random_state	42	Reproducibility — same result bar bar

PART 3 — Algorithms & Metrics Deep Dive

3.1 K-Means Clustering

Ki kore? k-ta cluster center (centroid) random-e place kore, tahole iterate kore — har document take tar sabase kache centroid-er cluster-e dey, tahole centroids re-calculate kore. Jokhan cluster change hoya bondho hoy, shesh.

> **Amra k=5 use korechhi keno?** Elbow Method — WCSS (Within-Cluster Sum of Squares) k=1 theke k=10 plot korle k=5 te ek clear 'bend' (elbow) dekh■ jay. Tar pore WCSS reduction matro 8% — tai k=5 optimal.

- **Result:** 5 ta well-defined cluster — Competition Law, IP & Contracts, Tax & Corporate, Immigration, Appellate
- **Speed:** 1,000 document-e 2 second-er kom time-e converge kore
- **DB Index:** 0.8477 — tin algorithm-er modhe shobcheye bhalo (lower = better)
- **Silhouette:** 0.3786 — moderate cluster separation

3.2 HDBSCAN

Ki kore? Density-based clustering. Bole — 'jei areas-e documents ghono ghono (dense), sei areas cluster, baki gulo noise.' Hierarchy banay different density level-e.

- **Automatic clusters:** k manually set kora lagena — algorithm nijei decide kore
- **Noise handling:** ~23 ta document 'noise' mark hoy (kono cluster-e fit kore na)
- **Result:** 60 micro-clusters — fine-grained sub-topics capture kore
- **Silhouette:** 0.4968 — tin algorithm-er modhe shobcheye beshi (better!)
- **Problem:** 60 cluster practical noy high-level organization-er jonno

> **minPts = 5** use kor■ hoyeche — ek cluster-e minimum 5 document thakte hobe. Ei parameter control kore cluster-er minimum size.

3.3 Agglomerative Hierarchical Clustering (Ward's Linkage)

Ki kore? Shuru-te har document alag alag cluster. Tahole boro boro similar pair merge kore jabe, jotokhon 5 cluster na hoy. Ward's linkage use kore — merge korar time within-cluster variance minimum rakhe.

- **Dendrogram:** Ek tree-er moto diagram — kono document kotokhon similar tar history dekhay
- **k=5 cut:** Dendrogram-ke 5 ta major branch-e cut kora hoy
- **Immigration branch:** Shobcheye alag ar door — Immigration-er vocabulary unique
- **Tax & Competition:** Nearest neighbors in dendrogram — shared financial vocabulary
- **Silhouette:** 0.3348, DB Index: 0.8765

3.4 Evaluation Metrics — Quality Measure Kora

Silhouette Score (Higher = Better, range: -1 to +1)

Ki measure kore? Har document ki tar nija cluster-e well-fit? Ki kache kache similar documents, ki door door different cluster?

s(i) = (b(i) - a(i)) / max(a(i), b(i))
a(i) = mean distance to same cluster documents (intra-cluster)
b(i) = mean distance to nearest other cluster (inter-cluster)
Score > 0.5 = Strong clustering | 0.25–0.5 = Moderate | < 0.25 = Weak

Davies-Bouldin Index (Lower = Better, 0 = Perfect)

Ki measure kore? Within-cluster scatter vs between-cluster separation-er ratio. Cluster gulo ki compact ar ekta theke arekta ki door?

Algorithm	Silhouette ↑	DB Index ↓	Clusters	Noise	Best For
K-Means	0.3786	0.8477 ✓	5	0	Production routing
HDBSCAN	0.4968 ✓	N/A	60	~23	Exploratory analysis
Hierarchical	0.3348	0.8765	5	0	Audit / compliance

PART 4 — Results & Key Numbers to Memorize for Viva!

4.1 UMAP-er Magic — The Most Important Finding

UMAP use korar age vs porey — ei difference-ta paper-er shobcheye important finding:

Raw TF-IDF Silhouette	UMAP 2D Silhouette	Improvement	DB Index Improvement
0.1379	**0.3786**	**+174%**	**-64%**
High-dim clustering	After UMAP	Silhouette Score	0.85 vs 2.36

Viva-te ei number ta mone rakhbo: UMAP use korle K-Means Silhouette Score 174.5% improve hoy. Raw TF-IDF-e 0.1379 chilo, UMAP-er pore 0.3786 hoyeche. DB Index 2.3559 theke 0.8477 — 64% reduction.

4.2 5 Cluster — Ki Ki Discover Kora Holo?

K-Means automatically 5 ta Federal Court division discover koreche — kono expert input chara:

C1	**Competition Law** Keywords: power, cost, commiss, consum, competit	**Size ~210**
C2	**IP & Contracts** Keywords: mark, contract, claim, trade, patent	**Size ~198**
C3	**Tax & Corporate** Keywords: commission, taxat, compani, privileg	**Size ~205**
C4	**Immigration** Keywords: tribun, minist, immigr, multicultur	**Size ~187**
C5	**Appellate Procedure** Keywords: appeal, applic, discret, feder, rule	**Size ~200**

4.3 Cluster-er Interesting Insights

- **Immigration (C4)** shobcheye compact ar isolated — vocabulary ekodom unique (tribun, immigr, multicultur)
- **Cited & Referred To** shobcheye overlap kore — semantic-e similar (kono strong endorsement nai dono-tei)
- **Competition & Appellate** overlap kore — competition law appeals-e procedural language use hoy
- **Tax & Competition** dendrogram-e nearest neighbors — shared financial/regulatory vocabulary
- **IP & Contracts (C2)** 'Applied' outcomes beshi — doctrine-driven nature of IP law

PART 5 — 30 Likely Viva Questions with Full Answers

Section A — Basic Concept Questions (Q1–Q8)

1 Q: Ei paper-er main objective ki chilo?

Answer: Amader main objective chilo 1,000 ti Australian Federal Court-er legal case document ke automatically cluster kora — kono labeled training data chara — shudhumatro text-er mathematical representation use kore. Amra unsupervised machine learning pipeline design korechi jeta TF-IDF, UMAP, ebong tin ti clustering algorithm use kore.

■ *Shurute ei ek sentence bolo — 'We proposed an end-to-end unsupervised NLP pipeline for clustering 1,000 real Australian legal documents without any labeled data.'*

2 Q: Unsupervised learning mane ki? Supervised-er sathe parta?

Answer: Supervised learning-e labeled data lage — machine ke batiye dite hoy 'eita tax case, eita immigration case'. Unsupervised learning-e kono label nai — machine nijei pattern khuje ber kore. Legal domain-e labeled data expensive (annotator costs \$150-\$350/hour), tai unsupervised approach practical.

■ *Real life example: Google Photos jokhan tomar face automatically group kore — tumi kono naam bolo ni — ei holo unsupervised!*

3 Q: TF-IDF ki? Keno use korechho?

Answer: TF-IDF = Term Frequency × Inverse Document Frequency. TF mane kono word sei document-e kotobar ache. IDF mane overall corpus-e word ta koto rare. Multiply korle pawa jay — word ta ki sei document-er jonno specifically important? High TF-IDF mane word ta ei document-e common but overall rare — tai discriminative.

■ *Example: 'tribunal' word Immigration cases-e beshi, other cases-e kam — tai itar TF-IDF score Immigration cluster-e beshi hobe.*

4 Q: UMAP ki? Keno PCA ba t-SNE use koroni?

Answer: UMAP = Uniform Manifold Approximation and Projection. High-dimensional data ke low-dimensional-e project kore — both local AND global structure preserve kore. PCA linear, global structure preserve kore but local structure miss kore. t-SNE local structure bhalo rakh■ but $O(n^2)$ — slow for large data. UMAP $O(n \log n)$ — faster, better structure preservation.

■ *Key point: 1,175D TF-IDF vector ke 2D-te convert kora — jate similar documents 2D space-e kache kache thake.*

5 Q: Curse of Dimensionality mane ki?

Answer: High-dimensional space-e shob pairwise distance almost equal hoye jay — cluster boundary distinguish kora impossible hoye jay. Imagine koro 1,175 different axis-e points — practically shob point 'same distance apart' mone hoy. Etar jonno raw TF-IDF-e clustering kaaj kore na bhalo — amader Silhouette Score chilo matro 0.14.

■ *UMAP eitar solution — dimension reduce kore cluster-able space taire kore.*

6 Q: Tomar dataset-e kototai document chilo? Keno ei number?

Answer: 1,000 documents — 5 category theke exactly 200 kore balanced stratified sampling. Original dataset-e 24,985 cases chilo but severely imbalanced — Cited category chilo 48.9% (12,219 cases) kintu Approved chilo matro 0.4% (108 cases). Balanced sampling ensure kore je kono category dominant na hok, fair comparison hok.

■ *Viva te bolbe: 'We used balanced stratified sampling to mitigate class imbalance which would bias clustering results.'*

7 Q: Preprocessing-e keno first 500 character-i use korecho puro text na?

Answer: Australian Federal Court judgments-e party names, jurisdiction, ar case type prothom paragraph-e thake. Ei opening sentences shobcheye discriminative content carry kore. Puro text nile noise beshi, computation time beshi, ar important features dilute hoye jay. 500 characters e tumi case-er 'identity' dhore fel■■ paro.

■ *Trade-off acknowledge korbe: 'This is a limitation — we miss argumentative body and legal reasoning, which future work should address.'*

8 Q: Vocabulary-te kototai term chilo?

Answer: Raw unique tokens: 14,823. Minimum document frequency filter (≥5 documents) apply korle: 3,241. Word length filter [3–20 characters] apply korle final vocabulary: 1,175 terms. Matrix dimensions: 1000 × 1175 with 94.7% sparsity — meaning most entries are zero.

■ *Sparsity 94.7% mane 95% jagay zero — eijonno efficient sparse matrix representation use kora hoy.*

Section B — Algorithm & Technical Questions (Q9–Q18)

9 Q: K-Means algorithm ki? Step-by-step explain koro.

Answer: K-Means: (1) k=5 random centroid place koro. (2) Har document take nearest centroid-er cluster-e assign koro. (3) Har cluster-er mean calculate kore new centroid baro. (4) Step 2-3 repeat koro jotokhon cluster change na hoy. Amra K-Means++ initialization use korechhi better convergence-er jonno, nstart=50, iter.max=300.

■ *Formula: $J = \Sigma \Sigma ||x_i - \mu_k||^2$ — minimize koro total within-cluster sum of squares.*

10 Q: Keno k=5 choose korecho? Koto howa uchit chilo?

Answer: Elbow Method use korechi — WCSS (Within-Cluster Sum of Squares) k=1 theke k=10 plot korle k=5 te ek clear elbow/inflection point dekh■ jay. k=5 theke k=6-e gele WCSS reduction drop kore 8%-er niche — marginal gain negligible. Also, dataset-e 5 ti ground-truth categories achhe.

■ *Always bolbe: 'k=5 is both statistically justified via the Elbow Method and semantically consistent with the 5 citation outcome categories in our dataset.'*

11 Q: HDBSCAN K-Means theke kotokhon better? Keno production-e K-Means recommend korechho?

Answer: HDBSCAN Silhouette Score-e better (0.4968 vs 0.3786) — density-based micro-clusters more cohesive. But HDBSCAN 60 clusters produce korche ar ~23 documents noise-e fel■che — high-level document organization-er jonno impractical. K-Means matro 5 interpretable cluster diyeche, zero noise, best DB Index (0.8477), ar 2 second-e converge koreche.

■ *Recommendation: 'K-Means for production routing, HDBSCAN for exploratory fine-grained analysis.'*

12 Q: Ward's Linkage mane ki? Hierarchical clustering-e keno Ward's use korechho?

Answer: Ward's linkage criterion: merge step-e total within-cluster variance increase minimize kore. Formula: $d_{Ward}(A,B) = \sqrt{(2n_An_B/(n_A+n_B))} \times ||\mu_A - \mu_B||$. Eita compact, equally-sized clusters produce kore — legal document organization-er jonno ideal. Complete linkage ba single linkage chained clusters produce korte pare jo legal taxonomy-er jonno suitable noy.

■ *Ward's linkage produces the most interpretable, compact clusters for this application.*

13 Q: Silhouette Score formula explain koro.

Answer: $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$. $a(i)$ = mean distance to all other documents in same cluster (intra-cluster distance). $b(i)$ = mean distance to all documents in nearest other cluster (inter-cluster). Score range: -1 to +1. +1 = perfect clustering, 0 = overlapping, -1 = wrong cluster. > 0.5 = strong, 0.25-0.5 = moderate, < 0.25 = weak.

■ *Amader K-Means Silhouette 0.3786 — moderate but meaningful structure. HDBSCAN 0.4968 — approaching strong.*

14 Q: Davies-Bouldin Index ki measure kore?

Answer: DB Index = (1/k) × Σ max(j≠i) [(si + sj) / dij]. si = mean distance from cluster i documents to centroid. dij = distance between centroids i and j. Eta within-cluster scatter ar between-cluster separation-er ratio. Lower DB = better clustering. 0 = perfect. K-Means = 0.8477 (best), Hierarchical = 0.8765.

■ *Ek line-e: 'DB Index measures how compact clusters are relative to their separation from each other.'*

15 Q: Porter Stemming mane ki? Example dao.

Answer: Porter Stemming morphological variants ke root form-e reduce kore — jate different forms of same word ek sathe count hoy. Example: agreements → agre, immigration → immigr, tribunal → tribun, applying → appli. Ei way, vocabulary size kame ar related terms ek group-e pore.

■ *Limitation: over-stemming hoye jete pare — 'computer' ar 'computation' ek stem-e pore jete pare, kinto meaning different.*

16 Q: Cosine similarity keno Euclidean distance-er bajaye use korechho UMAP-e?

Answer: Legal documents length-e vary kore — chhoto summary theke boro judgment. Euclidean distance document length diye influence hoy — lamba document mane shob direction-e beshi value. Cosine similarity shudhu direction measure kore, magnitude noy — tai document length-er effect avoid kore. Eijonno legal text clustering-e cosine preferred.

■ *Cosine similarity = dot product / (magnitude A × magnitude B) — 1 = same direction = same content.*

17 Q: HDBSCAN-er noise points mane ki?

Answer: HDBSCAN-e ~23 document 'noise' mark hoyeche — mane ei documents kono dense region-er part noy, kono cluster-e properly fit kore na. Ei documents boundary cases — maybe language mixed ba ambiguous content ache. HDBSCAN-er advantage holo je eta force kore na shob document ke kono cluster-e dhuke dite.

■ *Noise points = outliers = documents that don't clearly belong to any practice area.*

18 Q: Tomar pipeline-er shobar cheye important methodological contribution ki?

Answer: UMAP preprocessing-er empirical demonstration. Raw TF-IDF clustering Silhouette = 0.14 (well below 0.25 threshold for meaningful structure). UMAP projection-er pore K-Means Silhouette = 0.38, HDBSCAN = 0.50 — 174% improvement. Ei finding future legal NLP studies-ke inform korbe — distance-based clustering-er age UMAP use kora essential.

■ *Bolbe: 'This paper is the first to demonstrate UMAP's critical role as a preprocessing step for legal document clustering.'*

Section C — Results & Interpretation Questions (Q19–Q24)

19 Q: Tin algorithm-er modhe konta best? Kono single answer ache?

Answer: Kono single 'best' nei — depends on use case. HDBSCAN best Silhouette (0.4968) — fine-grained analysis. K-Means best DB Index (0.8477) ar most interpretable 5 clusters — production routing. Hierarchical best explainability via dendrogram — regulatory audit trails. Amader recommendation: K-Means for primary production, HDBSCAN for exploratory research.

■ *Viva-te nuanced answer dao — examiner prefer kore jocheye candidates who understand trade-offs.*

20 Q: Immigration cluster keno shobcheye compact ar isolated?

Answer: Immigration cases-er vocabulary ekodom unique ar distinctive — tribun, immigr, multicultur, minist — ei words other legal domains-e almost appear kore na. Other categories (Competition, Tax, Appellate) share kore common legal procedural vocabulary. Ei linguistic uniqueness Immigration cluster-er Silhouette Width highest (≈0.58) kore.

■ *In UMAP plot, Immigration cluster is most visually compact and separated from others.*

21 Q: Competition Law ar Appellate Procedure keno overlap kore?

Answer: Competition law appeals frequently use procedural/appellate language — 'appeal', 'discretion', 'application'. Dono cluster-e similar vocabulary thakay boundary ambiguous. Ei holo expected finding — real-world-e court proceedings often cross-domain language use kore. Silhouette analysis-e Clusters 1 ar 5 most negative values dekha jay.

■ *Ei answer-e bolbe: 'This overlap is semantically meaningful, not a flaw in our clustering — it reflects real-world linguistic ambiguity.'*

22 Q: 5 cluster ki Federal Court-er official divisions-er sathe match kore?

Answer: Hya — perfectly! Amader 5 ta cluster map kore onto: (C1) Competition & Consumer Division, (C2) Intellectual Property Division, (C3) Corporations & Taxation Division, (C4) Administrative & Constitutional Law Division (Immigration), (C5) General & Commercial Division (Appellate). Kono domain expert input chara automatic discovery — eta paper-er key validation.

■ *Bolbe: 'Unsupervised discovery of official court divisions validates the discriminative power of TF-IDF features in the legal domain.'*

23 Q: Cluster Distribution Analysis theke ki interesting pattern dekhecho?

Answer: C4 (Immigration) 'Considered' ar 'Followed' outcomes beshi — immigration tribunal-e prior decisions carefully engage kora hoy. C2 (IP & Contracts) 'Applied' outcomes beshi — IP law doctrine-driven, principles directly apply hoy. C1 ar C5 (Competition & Appellate) most mixed distributions — linguistic boundary overlap consistent.

■ *Cluster purity analysis shows that unsupervised clusters correlate with ground-truth citation outcomes.*

24 Q: Paper-er limitations ki ki?

Answer: 4 ta main limitation: (1) Text truncation — first 500 char only, argumentative body miss kora hoy. (2) TF-IDF bag-of-words — word order, syntax, semantics ignore kore ('not liable' vs 'liable' same treat hoy). (3) Balanced sampling — real-world imbalance (Cited 48.9%) reflect kore na. (4) English only — multilingual jurisdictions cover kore na.

■ *Limitations honest-ly acknowledge kora good academic practice — examiner appreciate kore.*

Section D — Application & Future Work Questions (Q25–Q30)

25 Q: Ei research-er practical applications ki?

Answer: 5 ta direct application: (1) Judicial Case Management — incoming filings automatically route kora appropriate Federal Court division-e. (2) Legal E-Discovery — large document productions-e cluster-based prioritization. (3) Precedent Discovery — new case-er cluster query kore relevant prior cases find kora. (4) Legal Taxonomy Construction — automatic domain taxonomy generation. (5) Academic Baseline — future legal NLP research-er reproducible baseline.

■ *Real impact: Legal practitioners spend 30-40% of billable hours on document review — automated clustering dramatically reduces this.*

26 Q: Future work-e ki ki improve kora jete pare?

Answer: 5 ta direction: (1) LEGAL-BERT ba LexLM use kore contextual embeddings — TF-IDF er chey■ semantics-aware. (2) Semi-supervised clustering — partial citation labels use kore constrained K-Means. (3) Temporal analysis — decades-er Federal Court language evolution study. (4) Full corpus scaling — 24,985 cases-e approximate nearest-neighbor UMAP. (5) Cross-jurisdiction — Australian, US, UK court comparison.

■ *Bolbe: 'The most impactful next step would be replacing TF-IDF with LEGAL-BERT embeddings to capture legal reasoning semantics.'*

27 Q: LEGAL-BERT ki? TF-IDF theke kono better?

Answer: LEGAL-BERT holo BERT-er domain-adapted version — 12 GB diverse legal text-e pretrained (US court opinions, contracts, legislation). Contextual embeddings generate kore — same word different context-e different vector. 'Consideration' (contract law) vs 'consideration' (general) — LEGAL-BERT distinguish korte pare, TF-IDF parena. Kintu LEGAL-BERT computational resources beshi lage ar labeled fine-tuning data dorkar.

■ *TF-IDF is our baseline — LEGAL-BERT would be the improvement direction for future work.*

28 Q: Tomar paper-e kono statistical significance test kora hoyeche?

Answer: No — amader paper qualitative metrics (Silhouette Score, DB Index) use koreche clustering evaluation-er jonno. Unsupervised learning-e statistical significance testing standard practice noy karon ground truth labels nai. HDBSCAN-er Relative Validity Index alternative metric provide kore but fair comparison-er jonno non-noise points-er Silhouette report korechi.

■ *Honest answer — 'Statistical significance testing is a limitation; future work could use bootstrap resampling for confidence intervals on metrics.'*

29 Q: Tomar research ki industry-te deploy kora jabe? Kono challenges achhe?

Answer: Yes, deployable! K-Means model 2 second-e 1,000 document cluster korte pare. Production challenges: (1) Real imbalance handle kora (Cited 48.9%). (2) New documents streaming classification. (3) Model retraining jokhan court new divisions add kore. (4) Explainability — judge ba lawyer ke explain kor■■ hobe keno ek document ei cluster-e. Hierarchical clustering-er dendrogram explainability-te help korte pare.

■ *Deployment-ready pipeline with R 4.5.3 — complete reproducible code.*

30 Q: Ekta sentence-e tomar paper-er contribution explain koro.

Answer: We demonstrated that combining domain-specific TF-IDF preprocessing, UMAP dimensionality reduction, and multi-algorithm clustering — without any labeled data — can automatically discover the five official practice divisions of the Australian Federal Court from 1,000 real case documents, with UMAP improving clustering quality by 174% over raw TF-IDF, providing a scalable, interpretable, label-free solution for legal document organization.

■ *Ei sentence-ta mone rakhba — viva-r shuru ba shesh-e use korbe. Confidence-er sathe bolo!*

Quick Reference Cheat Sheet — Defense-er Age Ek Bar Dekho!

Nicher numbers gulo mone rakhle viva-te confident answer dite parbe:

Documents	Vocabulary	UMAP Dims
1,000	**1,175**	**2D**
5 × 200 balanced	after filtering	from 1,175D

K-Means Sil.	HDBSCAN Sil.	Hier. Sil.
0.3786	**0.4968 ✓**	**0.3348**
DB: 0.8477 ✓	60 clusters	DB: 0.8765

Raw TF-IDF Sil.	UMAP Improvement	DB Improvement
0.1379	**+174%**	**-64%**
Before UMAP	Silhouette Score	2.36 → 0.85

Algorithm	Best At	Use When
K-Means (k=5)	DB Index (0.8477) + Interpretability	Production document routing
HDBSCAN	Silhouette (0.4968) + Auto k	Exploratory fine-grained analysis
Hierarchical	Dendrogram explainability	Regulatory audit trails

Tomar Defense-er Jonno Best Wishes!

Mone rekho: Tumi ei paper-ta lekhechi — tumi ei system build korechhi. Confident thako, numbers mone rakho, ar limitations honest-ly accept koro. Examiner dekh■■ chai tumi ki *bujhechi* — shudhu memorize na. Good luck! ■